The Organizational Context – The Value of Data

From analytics to action

Today

- Learning objective #3
- Define the project's (commercial) value
- How do you define a project's value?
- Start with thinking about data and their value. What is the role of data in a specific organization? What is their source of value?

- <u>Concepts:</u>
- Data as a tool
- Data as a commodity
- Data as a practice
- Data as algorithmic intelligence

Teaching you two things at a time

- How to analyze an organizational context **using the concept of data value**
- Understanding the use of social science concepts in relation to an organizational problem

NB: Understanding the organization as context for data projects is key to understanding how to put analytics into action

Time to reassess data value: The many faces of data in organizations (Xu et al., 2024: 2-3) (1/2)

- Article address the following **research question**: In what ways can data contribute to business objectives, and how can we conceptualize the role of data in organizations?

- **Comprehensive literature review** – focus on the diverse and up-to-date use of data in organizations

- Keyword search of databases returned 3,508 academic publications:
  - Keywords: “data business value or strategic value”“data value or valuation”,“data (or analytics) and performance (or impact)”, “data monetization”,“data commodity”, and “data marketplace”
  - Main database: Web of Science, supplemented by AISeL and some journal websites incl. MIS Quarterly and The Journal of Strategic Information Systems
  - Final sample: 142 papers from 41 peer-reviewed journals and 8 premier conference proceedings

Time to reassess data value: The many faces of data in organizations (Xu et al., 2024) (2/2)

- **Contributions:**
  - Developed **typology** that shows the diverse roles that data play in organizations, demonstrating the distinctive characteristics associated with each role.
  - Discussion of how this typology can **assist practitioners** in leveraging this deeper understanding as they seek to effectively monetize, manage, and govern data assets in their organizations

- **Data exhibit varied characteristics depending on their usage** in different business contexts

- Primary roles that data play in organizations: **data as a tool, data as a commodity, data as a practice, and data as algorithmic intelligence**

1. **Data as tool** (Xu et al., 2024)

- Data are used to **support decision-making processes** in managerial contexts
- Attention is directed to “instrumental effects” of data and its role in mediation between individuals and environment.
- “While using data is more of a human centred activity (Knox, 2007), the user and the organization can vary the needs and purpose of using data, reshape the way of use, and improve the outcome (Orlikowski and Iacono, 2001; Jones, 2019).” p. 5
- In sum, data helps people make better decisions by turning information into managerial insight. Value comes through improving judgment, planning, and strategy.

2 examples: Data as tool

- **Retail demand forecasting**
  - A supermarket analyses historical sales data to decide how much inventory to order
  - Data does not “decide”; managers interpret dashboards and adjust decisions.
  - Value = better planning, reduced

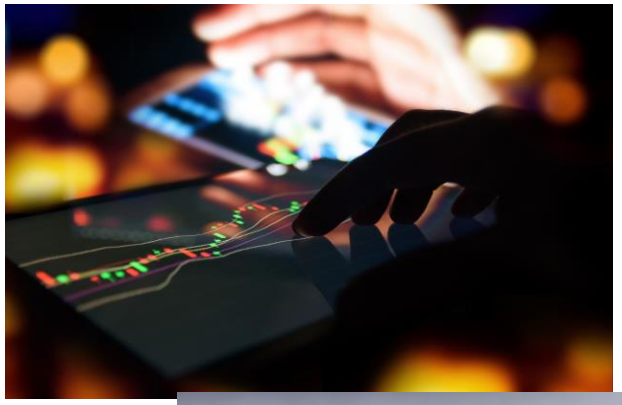
- **Hospital resource allocation**
  - Hospital administrators use patier data to allocate staff and beds
  - Data acts as a decision-support instrument

2. **Data as commodity** (Xu et al., 2024)

- Data is traded or sold as goods to **generate direct revenue or exchanged for other resources**, mostly between organizations.
- **Data trading** emerges as the main scene; the value creation process, mostly inter-organizational, involves a two-sided market coordination between buyers and sellers
- Common characteristic is the attribution of a **monetary value during transactions**

2 examples: Data as commodity

- **Consumer data marketplaces**
  - Platforms sell aggregated consumer behaviour datasets to advertisers
  - Data becomes a product in a marketplace



- **Weather data services**
  - Companies purchase climate datasets to optimize logistics or agriculture

3. **Data as practice** (Xu et al., 2024)

- Repeatable patterns of activities that **leverage data for continuous improvements in business operations**, and for optimizations of products or services
- Significance of data **in driving digital innovations** and **facilitating organizational learning**
- Ensemble view emphasizes the **interactions and relationships between IT artifacts and the groups involved in their construction**, implementation, and use

2 examples: Data as practice

- **Recommendation systems**
  - Streaming services like **Netflix** continuously learn from user interactions to improve recommendations
  - Data is not a one-time resource; it's embedded in everyday operations

- **Ride-sharing platforms**
  - Companies like **Uber** use trip data to refine pricing, routing, and service availability
  - Learning occurs through repeated feedback cycles

- Important: Focus on **organizational routines** and "learning-by-doing." Data value emerges through repeated practice, not single decisions.

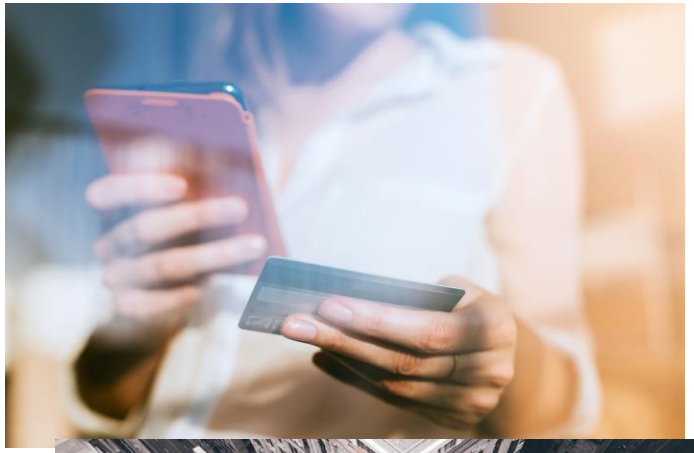
4. Data as algorithmic intelligence (Xu et al., 2024)

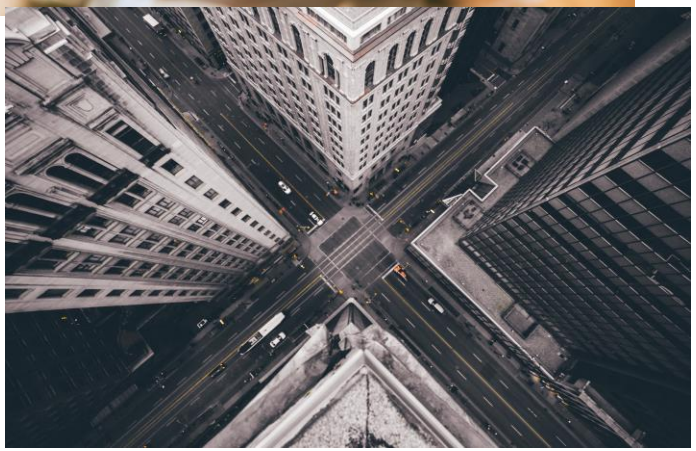
- Role of data in **supporting process automation and algorithmic decision-making**, thereby generating value for organizations at an unprecedented pace and with remarkable versatility, potentially surpassing our imaginations. This is achieved through various forms of artificial intelligence (hereafter referred to as “AI”) and machine learning applications in business.

- The computational view: Technology can be represented here through **algorithmic endeavours to build new or enhance existing computational systems that can support some human activity**.

2 examples: Data as algorithmic intelligence

- **Fraud detection**
  - Banks use machine learning models that automatically flag suspicious transactions
- **Predictive maintenance**
  - Industrial sensors feed models that autonomously trigger maintenance alerts
- Important: algorithms become active decision actors rather than passive tools (**shifting agency)**





Summary

Data as …	Primary question	Who drives value?
Tool	What should we decide?	Humans
Commodity	What can we sell?	Market actors
Practice	How do we continuously improve?	Organization and routines
Algorithmic intelligence	What can be automated?	Algorithms

Implications for practice (Xu et al., 2024)

1. **Data as tool:** emphasis on individuals and that organizations develop their data related skills and competencies (e.g., data literacy).
2. **Data as comodity:** Data should no longer been seen as a cost for organizations but a source of profit. Data as most liquid assets that an organization possesses.
3. **Data as practice:** Importance of embedding data within a repeatable process and formalizing it. People are just one part of this ensemble (e.g., centralized data platforms to democratize enterprise data to a broader audience).
4. **Data as algorithmic intelligence:** Potential of data in replacing human labour and enhancing our ability in solving complex problems.

Exercise (for time reasons in reading sessions)

Please take one of the examples presented (e.g., recommender systems) and discuss and show how data plays **all four roles simultaneously!**

DTU